
Training Dynamics: AoI (↓ better) vs Decode Rate (↑ better)
Average AoI (solid lines)
Decode Rate % (dashed lines)
Training Episode
28
27
26
25
24
23
22
21
20
19
25
20
15
10
5
0
0
1,000
2,000
3,000
4,000
5,000
Proposed (PPO+SAC, β_{min}=0.2)
Ideal RIS (β_{min}=1.0, Upper Bound)
PPO-Only (No RIS Optimization)
Random Baseline
AoI (solid)
Decode Rate (dashed)